higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T1390**(E)**(A12)T

NATIONAL CERTIFICATE

POWER MACHINES N5

(8190035)

12 August 2019 (X-Paper)
09:00–12:00

REQUIREMENTS: **Steam Tables (BOE 173)**
Superheated Steam Tables (Appendix to BOE 173)

Candidates need drawing instruments.

Nonprogrammable calculators may be used.

This question paper consists of 5 pages, a formula sheet of 3 pages and 1 answer sheet.

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NATIONAL CERTIFICATE
POWER MACHINES N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. QUESTION 5.2 should be answered on the ANSWER SHEET (attached). Write your EXAMINATION NUMBER in the space provided, detach it from the question paper and hand it in with the ANSWER BOOK.
6. Use only BLUE or BLACK ink.
7. Write neatly and legibly.

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QUESTION 1

1.1 Name THREE methods of firing a boiler. (3)

1.2 Name the TWO most common pumps that can be used to extract air in a condenser. (2)

1.3 Charles' law describes the temperature versus volume relationship when the pressure is constant.

Is the relationship between temperature and volume directly or inversely proportional? (1)

1.4 Define *isochronism* and *hunting* in terms of governors. (4)

[10]

QUESTION 2

0,245 kg of gas with a density of 1,2 kg/m^3 is compressed adiabatically from a pressure of 150 kPa to a pressure of 800 kPa. R = 0,293 kJ/kg.K and C_p = 1,005 kJ/kg.K.

Calculate the following:

2.1 The adiabatic index (3)

2.2 The original volume and temperature (6)

2.3 The final absolute temperature (4)

2.4 The work done in kJ/kg (4)

2.5 The change in internal energy (3)

[20]

QUESTION 3

Feedwater enters an evaporator at 44,8 °C. Wet steam with a dryness fraction of 0,85 enters the superheater at 1 800 kPa and the steam leaves at 250 °C. The equivalent evaporation forms and at 100 °C it is 12 kg of steam per kg fuel. The calorific value of the fuel is 32 MJ/kg and the specific heat capacity of superheated steam is 2,1 kJ/kg.K.

Calculate the following:

3.1 The mass of the steam per kg fuel (7)

3.2 The efficiency of the plant (5)

3.3 The heat transferred to the evaporator per kg of fuel (4)

3.4 The heat transferred to the superheater per kg of fuel (4)

[20]

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QUESTION 4

A surface condenser receives steam at a rate of 100 kg per minute. The vacuum reading is 670 mmHg and the barometer reading is 760 mmHg. The air pump temperature is 38 °C and the condensate temperature is 42,7 °C. The cooling water enters at 21,1 °C and leaves at 40 °C. The gas constant is 287 J/kg.K and the specific heat capacity of water is 4,187 kJ/kg °C.

Calculate the following:

4.1 The absolute pressure of the steam in the condenser (2)

4.2 The amount of cooling water required in kg per second (7)

4.3 The mass of air in kg in the condenser if the capacity is 2 m^3 (5)

4.4 The mass of air in kg if the capacity is increased by 0,5 m^3 (4)

4.5 By how many kilograms has the mass of air increased due to the rise in the capacity of air (2)

[20]

QUESTION 5

A nozzle inclined at an angle is 18° to a turbine's wheel of rotation, discharges a mass of steam at a rate of 15 kg/min. The turbine blades rotate at a radius of 222,5 mm about the turbine center of rotation and the turbine wheel rotates at a speed of 6 500 r/min. The blade's relative inlet angle is 25° and the blade's outlet relative velocity is 185 m/s. The steam is axially discharged to the plane of the wheel.

5.1 Determine the mean blade velocity. (3)

5.2 Use scale 1 cm = 30 m/s to construct a velocity diagram on the ANSWER SHEET (attached) in landscape. Indicate the velocity values in m/s on the diagram.

NO MARKS WILL BE AWARDED FOR THE VELOCITY DIAGRAM IF THE INSTRUCTIONS ARE NOT FOLLOWED.

Write your EXAMINATION NUMBER in the space provided on the ANSWER SHEET, detach it from the question paper and hand it in with the ANSWER BOOK. (10)

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5.3 Use the velocity diagram you constructed in QUESTION 5.2 to determine the following:

5.3.1 The velocity of the gas leaving the turbine

5.3.2 The relative blade outlet angle

(2 × 1) (2)
[15]

QUESTION 6

Butane (C_4H_{10}) is burnt with 35% excess air. The atomic mass of carbon is 12, hydrogen is 1, oxygen is 16 and nitrogen is 14. There is 1 kg of fuel.

Calculate the following:

6.1 The theoretical mass of oxygen to completely combust the 1 kg of fuel (7)

6.2 The mass of the products and percentages of combustion (8)
[15]

TOTAL: 100

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POWER MACHINES N5

FORMULA SHEET

1. $Q = W + \Delta U$
2. $\Delta U = mC_{\upsilon}\Delta T$
3. $Q = mC_{\upsilon}\Delta T$
4. $Q = mC_{p}\Delta T$
5. $Q = P_1V_1 \ln \frac{V_2}{V_1}$
6. $\Delta S = m\left(C_{\upsilon} \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1}\right)$
7. $W = P_1\Delta V$
8. $W = P_1V_1 \ln \frac{V_2}{V_1}$
9. $W = \frac{P_1V_1 - P_2V_2}{n-1}$
10. $W = \frac{P_1V_1 - P_2V_2}{\gamma - 1}$
11. $R = C_p - C_{\upsilon}$
12. $\gamma = \frac{C_p}{C_{\upsilon}}$
13. $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$
14. $PV = mRT$
15. $P_1V_1 = P_2V_2$
16. $P_1V_1^n = P_2V_2^n$
17. $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} = \left(\frac{V_1}{V_2}\right)^{n-1}$
18. $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$
19. $h = h_f + \chi h_{fg}$
20. $h = h_g + C_p\Delta T$
21. $h = h_f + h_{fg} = h_g$
22. $V_{\text{sup}} = \frac{n-1}{n}\left(\frac{h_{\text{sup}} - 1\,941}{P}\right)$
23. $\chi = \frac{V_m}{V_g}$
24. $\chi = \frac{M}{M + m}$
25. $U = H - PV$
26. $gZ_1 + U_1 + P_1V_1 + \frac{1}{2}C_1^2 + Q =$

    $gZ_2 + U_2 + P_2V_2 + \frac{1}{2}C_2{}^2 + W$
27. $\eta = \frac{\dot{m}_s(h_2 - h_1)}{\dot{m}_f\, CV}$
28. $EE = \frac{\dot{m}_s(h_2 - h_1)}{\dot{m}_f\ 2\,257}$
29. $p = (B_m \pm M_m)\frac{101{,}325}{760}$

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30. $m = \frac{100}{23}\left[C\frac{8}{3} + 8H_2 + S - O_2\right]$

31. $C_xH_y + \left(x + \frac{y}{4}\right)O_2 = xCO_2 + \frac{y}{2}H_2O$

32. $H.C.V. = (CV_C.C) + CV_{H_2}\left(H_2 - \frac{O_2}{8}\right) + (CV_s.S)$

33. $L.C.V. = H.C.V. - h_{fg}\ (9H_2)$

34. $H.C.V. = \frac{(m_w + m_e)\ C_p\Delta T}{m_f}$

35. $W = P_1V_e\left(\frac{n}{n-1}\right)\left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right] = mRT_1\left(\frac{n}{n-1}\right)\left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right]$

36. $\eta_c = \frac{V_e}{V_s}.100 = 1 - \frac{Vc}{Vs}\left[\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}} - 1\right] = 1 + \alpha - \alpha(r_p)^{\frac{1}{n}}$

37. $\eta_\alpha = \frac{V_\alpha}{V_s}.100$

38. $F_c = \dot{m}(C_{fe} - C_{fi})$

39. $P = \dot{m}U[C_{wi} - (-C_{we})]$

40. $\eta = \frac{2U[C_{wi} - (-C_{we})]}{C_{ai}{}^2}.100$

41. $U = \pi DN$

42. $\dot{m}V = AC$

43. $(m + M)g = m\omega^2h$

44. $Vs = \frac{\pi}{4}D^2L$

45. $\theta_1 = t_c - twi$

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46. $\theta_2 = t_c - two$

47. $Log.temp.diff. = \dfrac{\theta_1 - \theta_2}{\ell n \dfrac{\theta_1}{\theta_2}}$

48. $Piso = P_1 V_1 \ell n \left(\dfrac{P_2}{P_1} \right)$

49. $Pact = \dfrac{n}{n-1} P_1 V_1 \left[\left(\dfrac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$

50. $Niso = \dfrac{Piso}{Pact} \bullet 100$

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ANSWER SHEET

EXAMINATION NUMBER:

QUESTION 5.2

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